

Feeder channel
This channel is dry because the rock fracture has blocked the water supply.

rising steam and gases, rather than hot water, occupy some channels

Rock fracture
Caused by an earthquake, this fracture has broken the channel supplying hot water to a geyser above.

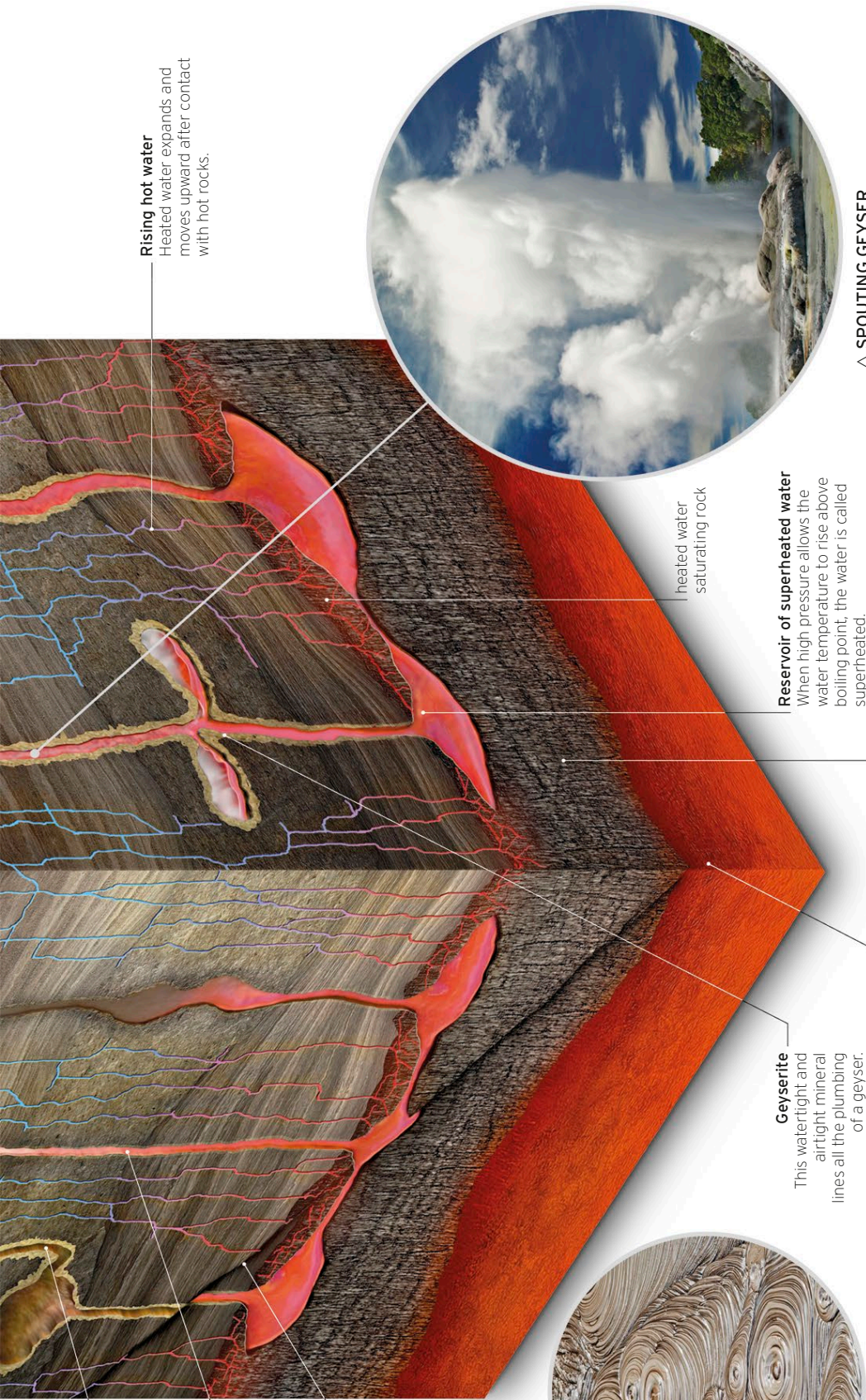
▷ MUD SWIRLS

Bubbling mud pools occur at several sites around Rotorua. As the bubbles burst, they create intricate swirling patterns on the mud surface.



▲ GEOTHERMAL FEATURES

This illustration shows a typical geothermal landscape that includes many of the features seen at Rotorua. The geothermal areas lie over volcanic calderas. These still have magma under them. The magma heats overlying rock, which in turn heats water that has percolated downward. This heated water supplies the geothermal features.



Rising hot water
Heated water expands and moves upward after contact with hot rocks.

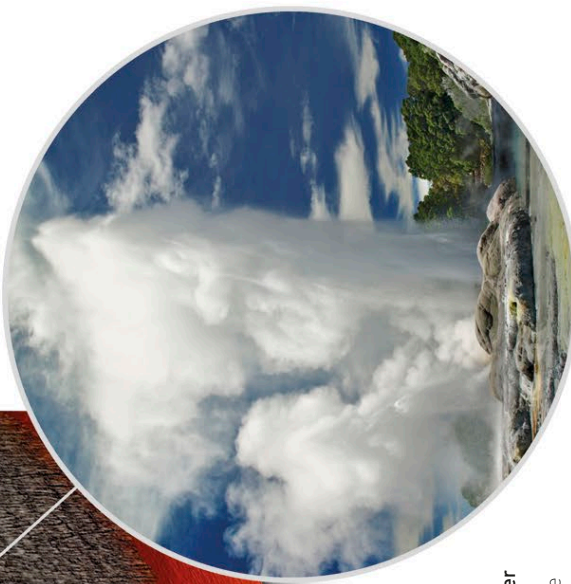
heated water saturating rock

Reservoir of superheated water
When high pressure allows the water temperature to rise above boiling point, the water is called superheated.

porous rock can hold large quantities of heated water

Geyserite
This watertight and airtight mineral lines all the plumbing of a geyser.

Hot rock
Underlying magma provides the energy that heats the overlying rock and powers the geothermal features.



△ SPOUTING GEYSER

On the outskirts of Rotorua City is a geyser field containing seven active geysers. The Pohutu geyser, shown here, produces the biggest and loudest eruptions.

The Pohutu geyser erupts up to 20 times a day, and eruptions can reach heights of up to 100 ft (30 m)

HOW A GEYSER ERUPTS

A geyser can form where hot water occupies a system of underground channels and chambers lined by a mineral called geyserite. There also needs to be a constriction in the channel to the surface. Superheated water flows up from below, increasing pressure in the system until the high pressure blasts water out from the constricted outlet. Pressure then drops underground, allowing some superheated water to flash into steam. As this expands, it sustains the eruption, which continues until pressure drops to zero. The cycle then starts again.

